

6 (a) Define *magnetic flux*.

flux and magnetic flux linkage is common. Magnetic field is insufficient as it describe a region and not quantify the strength.

NYJC 2022 9749/03/J2Prelim/22
The magnetic flux through a plane surface is the product of the component of the magnetic flux density normal to the surface and the area of the surface.

[2]

- (b) A uniform conducting bar XY is pulled horizontally across long parallel frictionless conducting guide rails by a light inextensible string. The string passes over a frictionless pulley and is attached to a hanging block. The guide rails are placed in a vertical magnetic field of uniform magnetic flux density as shown in Fig. 6.1.

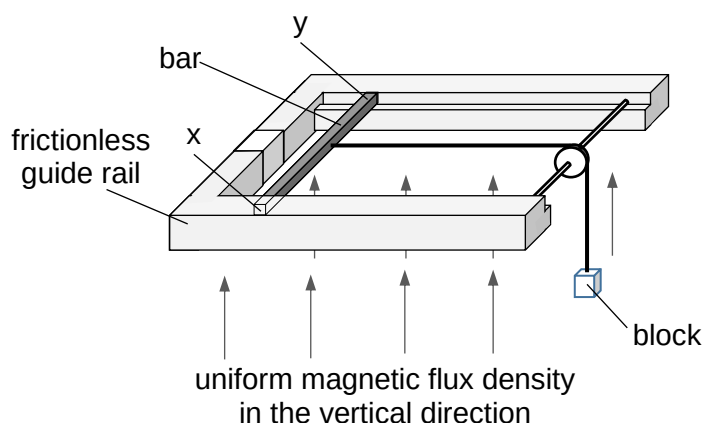


Fig. 6.1

The block is released from rest at time $t = 0$ s, and the bar starts to move.

- (i) An e.m.f. is induced in the bar. State which end X or Y of the bar is at a higher potential.

End X of the rod is at a higher potential.

Comment
Mainly guess work.

[1]

- (ii) Use Faraday's law and Lenz's law to explain the subsequent motion of the bar.

When the bar accelerates, the velocity increases and there is a change (increase) in magnetic flux linkage through the area enclosed by the rod and the conducting rails because the area is changing (increasing) OR cutting of flux/cutting of magnetic field lines. [B1]

By Faradays law, there is an e.m.f induced and by Lenz's law, a current is induced in a clockwise direction viewed from above to oppose the change in the magnetic flux linkage. [B1]

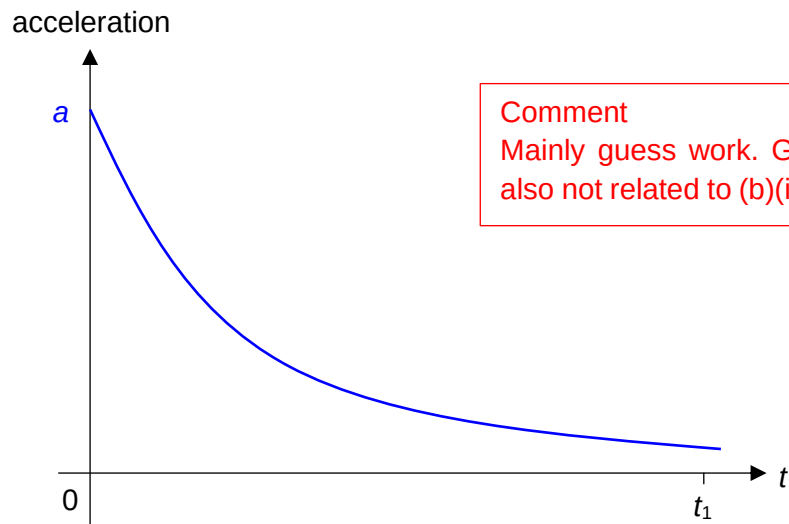
Hence by Fleming's left hand rule, a magnetic force is experienced by the bar towards the left, hence decreasing its acceleration/decreasing the net force acting on it. [B1]

Hence the speed of the bar increases at a decreasing rate. [B1]

Comment

Very poorly done. Majority failed to support change in flux linkage due to change in area of loop. Magnetic flux linkage is a scalar there should not be direction associated with it which is seen in many answers. Faraday's Law and Lenz Law are just stated without linking it to context of question. Faraday's Law do not account for induced current. Lenz Law (in this context) only used to oppose increase in magnetic flux linkage, it is incorrect to say that magnetic flux linkage is reduced. It should account for induced e.m.f. only. Many answers failed to describe direction of induced current adequately. Common misconception that reduction in net force result in deceleration when in fact it only leads to smaller acceleration. Many failed to describe motion of bar adequately to score.

On Fig. 6.2, sketch a graph to show the variation with time t of the acceleration of the rod from $t = 0$ to time $t = t_1$. Label clearly the acceleration at $t = 0$ as a .



Comment

Mainly guess work. Graph plotted are also not related to (b)(ii) when it should.

[B1] Shape of curve.

[B1] Start from a positive value, labelled a .

[2]

[Total: 6]